Ex:No: 5B Date: 01/04/26	Visualization and Classification of Iris Flower Species Using K-Nearest Neighbor Algorithm

Problem Description:

The Iris dataset is a well-known dataset in machine learning that contains measurements of iris flowers belonging to three different species: *Setosa*, *Versicolor*, and *Virginica*. Each flower is described using four numerical attributes: **Sepal Length, Sepal Width, Petal Length,** and **Petal Width.**

The objective is to visualize the dataset, apply the K-Nearest Neighbor (KNN) algorithm, and analyze the classification results.

Objective:

To visualize the Iris dataset and implement the K-Nearest Neighbor algorithm in R for classifying iris flower species based on their measurements.

Procedure:

- **Step 1: Install and Load Required Libraries**

  We install and load the necessary libraries for implementing the K-NN algorithm.

```
# Install necessary libraries
install.packages("class")
install.packages("caret")
install.packages("ggplot")
install.packages("GGally")
```

```
# Load the libraries
library(class)
library(caret)
library(ggplot)
library(GGally)
```

- **Step 2: Load and Prepare the Dataset**

  We use the iris dataset for classification and check its structure.

```
data(iris)
set.seed(123)
```

Normalize the numerical data (Feature Scaling)

```
normalize <- function(x) {
  return((x - min(x))/(max(x) - min(x)))
}
```

Apply normalization to numerical columns

```
iris_norm <- as.data.frame(lapply(iris[1:4],
                                  normalize))
iris_norm$species <- iris$species
```

- **Step 3: Split the Dataset into Training and Testing Sets**

  We split the dataset into 80% training and 20% testing.

```
split_index <- create DataPartition(iris_norm$species,
                                    p = 0.8, list = FALSE)
train_data <- iris_norm[split_index,]
test_data <- iris_norm[-split_index,]
```

Extract labels (target variable)

```
train_labels <- train_data$species
test_labels <- test_data$species
```

- **Step 4: Implement the K-Nearest Neighbor Algorithm**

  We use knn() function to build the model.

  k <- 5 # Choose number of neighbors

```
knn_predictions <- knn(train = train_data[, 1:4],
                       test = test_data[, 1:4],
                       cl = train_labels, k = k)
```

- **Step 5: Evaluate Model Performance**

  We compute accuracy using a confusion matrix.

```
conf_matrix <- confusionMatrix(knn_predictions,
                               test_labels)
print(conf_matrix)
```

- **Step 6: Visualize the Classification Results**

  We plot a scatter plot of Sepal.Length vs Sepal.Width with predicted class labels.

  # Scatter Plot: Sepal Length vs Sepal Width, colored by Species

```
ggplot(iris, aes(x = Sepal.Length, y = sepal.width,
                 color = Species)) +
  geom_point(size = 3) +
  labs(title = "Iris Dataset - Sepal Length vs Sepal
       Width",
       x = "Sepal Length", y = "Sepal Width") +
  theme_minimal()
```

OUTPUT:

Confusion Matrix and Statistics

Prediction	Reference Setosa	Versicolor	Virginica
Setosa	10	0	0
Versicolor	0	10	2
Virginica	0	0	8

Overall Statistics

Accuracy : 0.9333

95% CI : (0.7793, 0.9918)

No Information Rate : 0.3333

P-Value [Acc > NIR] : 8.747e-12

Kappa : 0.9

Mcnemar's Test P-value : NA

Statistics by class :

	class: setosa	class: versicolor	class: virginica
Sensitivity	1.0000	1.0000	0.8000
Specificity	1.0000	0.9000	1.0000
Pos Pred Value	1.0000	0.8333	1.0000
Neg Pred Value	1.0000	1.0000	0.9091
Prevalence	0.3333	0.3333	0.3333
Detection Rate	0.3333	0.3333	0.2667
Detection Prevalence	0.3333	0.4000	0.2667
Balanced Accuracy	1.0000	0.9500	0.9000

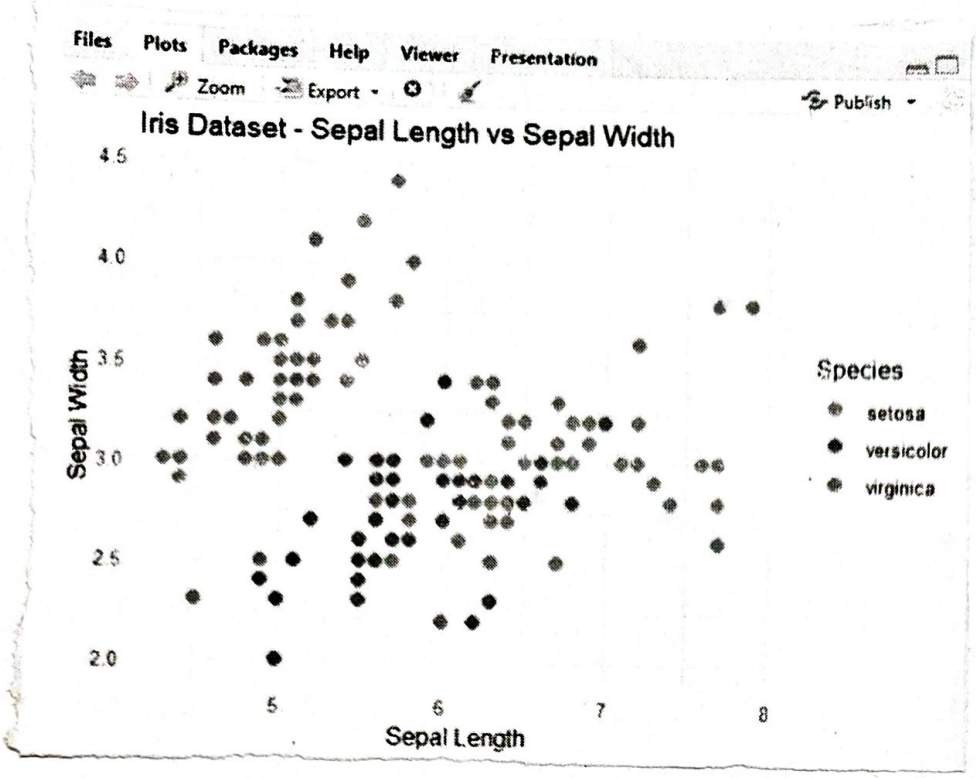
Now, we generate a pair plot using the ggpairs() function, which visualizes the relationships between all numeric features.

Pair Plot: Relationship between all numeric features

```
ggpairs(iris, aes(color = Species))
```

Result:

Thus, the program successfully demonstrates the visualization and classification of the Iris dataset using the K-Nearest Neighbor algorithm in R. The visualizations reveal clear class separations, and the KNN model achieves high classification accuracy, confirming its effectiveness for multi-class classification problems.


Files
Plots
Packages
Help
Viewer
Presentation
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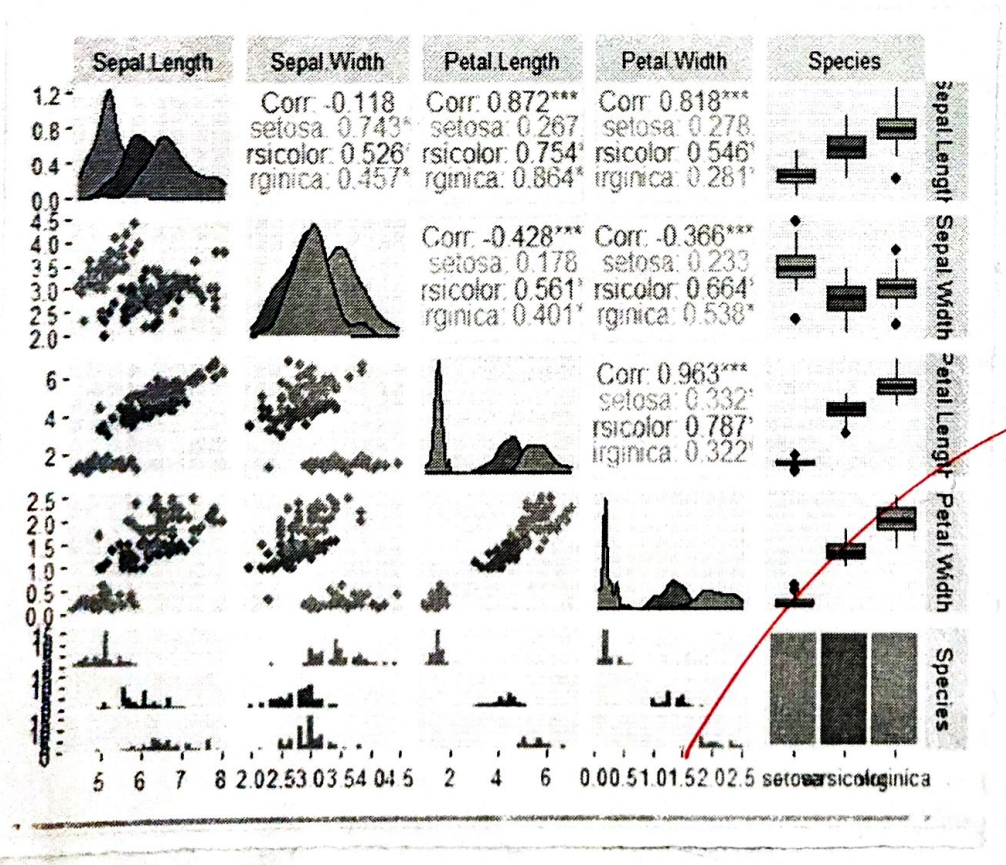
Export
Publish
Iris Dataset - Sepal Length vs Sepal Width
Sepal Width
Sepal Length
Species
setosa
versicolor
virginica


Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
Species
Corr: -0.118
setosa: 0.743*
rsicolor: 0.526*
rginica: 0.457*
Corr: 0.872***
setosa: 0.267
rsicolor: 0.754*
rginica: 0.864*
Corr: 0.818***
setosa: 0.278
rsicolor: 0.546*
irginica: 0.281*
Corr: -0.428***
setosa: 0.178
rsicolor: 0.561*
rginica: 0.401*
Corr: -0.366***
setosa: 0.233
rsicolor: 0.664*
rginica: 0.538*
Corr: 0.963***
setosa: 0.332*
rsicolor: 0.787*
irginica: 0.322*